JEE Mathematics: Quick Review with Grading and Practice Map - Determinant (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (66 questions; a question that appears in two of these is counted once): High = 6 or more questions, Medium = 1 to 5 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Introduction of Determinant

Question 01

Are you able to define the order of a determinant and how its elements are represented?

Model Answer

A determinant has the same number of rows and columns, written between two vertical bars; that number n is its order. a_{ij} is the element in row i and column j; $a_{11}, a_{22}, \ldots$ form the principal (leading) diagonal.

Ex: $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ has order 2 (4 elements); a third-order determinant has 9 elements and 9 minors.

Grade - Low

Practice map, main concept: none

Also used in: none

Value of a Determinant by Expansion

Question 02

How do you find the value of a determinant of order 2 by expansion?

Model Answer

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Ex: $\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} = 12 - 2 = 10.$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

How do you find the value of a determinant of order 3 by expansion?

Model Answer

Expand along a row/column with the sign pattern $+ - +$ / $- + -$ / $+ - +$ (element $\times$ its 2×2 minor):

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Ex: $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix} = 1(0 - 24) - 2(0 - 20) + 3(0 - 5) = 1.$

Grade - Medium

Practice map, main concept: Sheet: J-I 1 | DPP: 1(3, 5); 2(7, 17) | Question Bank: 113

Also used in: DPP: 1(2); 3(6)

Question 04

Along which row/column should we expand a determinant?

Model Answer

Along the row/column with the most zeros (the value is the same along any row or column). If there is none, first create zeros with row/column operations.

Ex: $\begin{vmatrix} 2 & 0 & 0 \\ 3 & 1 & 4 \\ 5 & 2 & 6 \end{vmatrix} = 2(6-8) = -4$ (expanded along R_1).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 1

Minor and Cofactor of element

Question 05

How do you find the minor and cofactor of any element of a determinant?

Model Answer

Minor M_{ij} = determinant left after deleting row i and column j; cofactor $C_{ij} = (-1)^{i+j} M_{ij}$.

Ex: in $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix}$: $M_{11} = 50 - 48 = 2$, $M_{12} = 40 - 42 = -2$, $C_{12} = 2$.

Grade - Low

Practice map, main concept: DPP: 1(6-7, 9)

Also used in: none

Question 06

Can you write the relation between minor and cofactor?

Model Answer

$$C_{ij} = (-1)^{i+j} M_{ij}$$

So $C_{ij} = M_{ij}$ when $i + j$ is even and $C_{ij} = -M_{ij}$ when $i + j$ is odd (sign chessboard $+ - +/ - + - / + - +$).

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-II 28

Question 07

What happens when all elements of a row/column are multiplied by their own cofactors and added?

Model Answer

The sum equals the value of the determinant (this is the expansion): $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = \Delta$, and likewise for any row or column.

Ex: in a 4×4 determinant $a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} + a_{14}c_{14} = \Delta$.

Grade - Low

Practice map, main concept: Sheet: J-II 28, 30 | DPP: 4(1)

Also used in: none

Question 08

What happens when all elements of a row/column are multiplied by the cofactors of another row/column and added?

Model Answer

The sum is zero: $a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23} = 0$ (it is a determinant with two identical rows).

Ex: in $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{vmatrix}$: $1 \cdot 4 + 2 \cdot (-22) + 3 \cdot 18 = 0$ (cofactors of R_2: $4, -22, 18$).

Grade - Low

Practice map, main concept: Sheet: J-II 26 | DPP: 1(8)

Also used in: Sheet: J-II 30

Sarrus Rule

Question 09

How do you use the Sarrus rule to expand a determinant of order 3?

Model Answer

Rewrite the first two columns to the right; add the three products along the down-right diagonals and subtract the three along the down-left diagonals (only for order 3):

$$D = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$$

Ex: $\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{vmatrix} = (0 + 0 + 40) - (15 + 0 + 24) = 1.$

Grade - Low

Practice map, main concept: none

Also used in: none

Properties of Determinant

Question 10

Can you write down all 6 properties?

Model Answer

P1: rows $\leftrightarrow$ columns (transpose): $D' = D$. P2: interchanging two rows (columns) changes the sign only. P3: two identical rows (columns) $\Rightarrow D = 0$ (also if proportional).

P4: multiplying one row (column) by k multiplies D by k. P5: if each element of a row (column) is a sum of two terms, D splits into a sum of two determinants. P6: if $D = f(x)$ and $f(a) = 0$, then $(x - a)$ is a factor of D.

Ex (P3): $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 5 & 7 & 9 \end{vmatrix} = 0$ (rows 1, 2 proportional).

Grade - Medium

Practice map, main concept: Sheet: J-I 4, 8; J-II 31, 33; C-I 21; C-II 14 | DPP: 1(4, 10-11); 2(5); 3(6); 4(7)

Also used in: Sheet: J-II 2, 10, 13; C-I 2 | DPP: 1(12); 2(9) | Question Bank: 95, 104

Question 11

How and when can we add two or more determinants without expanding?

Model Answer

Two determinants can be added when they are identical except in one row (column); add the elements of that row (column) and keep the rest (reverse of P5).

Ex: $\begin{vmatrix}1 & 2\\3 & 4\end{vmatrix}+\begin{vmatrix}5 & 2\\6 & 4\end{vmatrix}=\begin{vmatrix}6 & 2\\9 & 4\end{vmatrix}=6.$

Grade - Low

Practice map, main concept: Sheet: J-II 2, 11; C-I 2 | DPP: 2(14)

Also used in: Sheet: J-I 8

Question 12

When can we take sigma inside the determinant?

Model Answer

When only one row (column) contains the running index r and all other rows (columns) are free of r: sum that row (column) element by element, then evaluate.

Ex: $\sum_{r=1}^{n}\begin{vmatrix}r & 1\\2 & 3\end{vmatrix}=\begin{vmatrix}\sum r & n\\2 & 3\end{vmatrix}=\dfrac{n(3n-1)}{2}.$

Grade - Medium

Practice map, main concept: Sheet: J-II 14; J-M 20, 30 | DPP: 2(15); 3(8); 4(2, 8, 12)

Also used in: none

Question 13

What happens when all elements of a determinant are multiplied by k?

Model Answer

$$|kA| = k^n|A|$$

Here n is the order of the determinant.

Only one row multiplied by k gives kD; every row gives k^nD.

Ex: order 3, $D = 5$, all elements $\times 2$: new value $= 2^3 \cdot 5 = 40$.

Grade - Low

Practice map, main concept: Sheet: J-II 27 | DPP: 1(13)

Also used in: Sheet: J-II 33

Question 14

How do you identify a factor of a determinant without expanding?

Model Answer

Put $x = a$ (or $a = b$); if two rows (columns) become identical or proportional, $D = 0$ so $(x - a)$ is a factor. Compare degrees and one coefficient to fix the remaining constant.

Ex: $\begin{vmatrix}1 & 1 & 1\\a & b & c\\a^2 & b^2 & c^2\end{vmatrix}=0$ at $a = b$, $b = c$, $c = a$; degree 3 gives $D = (a-b)(b-c)(c-a)$.

Grade - Low

Practice map, main concept: Sheet: J-II 12; C-I 20 | DPP: 4(9)

Also used in: Sheet: C-II 9 | DPP: 3(3)

Row and Column Transformation

Question 15

How do you apply a given row/column transformation?

Model Answer

$R_i \to R_i + \lambda R_j$ means: replace row i by (row i) + $\lambda\times$(row j); only row i changes, all other rows are copied. Same for $C_i \to C_i + \lambda C_j$.

Ex: $\begin{vmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 10\end{vmatrix}$; $R_2 \to R_2 - 4R_1,\ R_3 \to R_3 - 7R_1$ gives $\begin{vmatrix}1 & 2 & 3\\ 0 & -3 & -6\\ 0 & -6 & -11\end{vmatrix} = 33 - 36 = -3.$

Grade - **High**

Practice map, main concept: Sheet: J-I 2, 7; J-II 1, 3, 5-6, 10, 13; C-I 10; C-II 9, 15 | DPP: 1(2, 12, 14-16); 2(4, 8-9, 19); 3(9); 4(3) | Question Bank: 99, 104

Also used in: Sheet: J-I 4-6; J-II 4, 8, 12, 31-32; J-M 8, 14, 31; J-A 3-4; C-I 6, 12, 15-17, 20-21; C-II 10, 16, 23-24 | DPP: 1(1, 5); 2(2, 18); 3(2, 4-5, 7, 11); 4(4, 7-8, 11) | Question Bank: 91, 101, 103, 106

Question 16

What happens to the value of a determinant under a row/column transformation?

Model Answer

$R_i \to R_i + \lambda R_j$: value unchanged. $R_i \leftrightarrow R_j$: sign changes. $R_i \to kR_i$: value $\times k$. $R_i \to xR_i + yR_j + zR_k$: value $\times x$.

Ex: $R_1 \to 2R_1 - R_2$ multiplies D by 2.

Grade - **Low**

Practice map, main concept: none

Also used in: none

Question 17

What should we keep in mind while applying more than one operation at a time?

Model Answer

Apply them one after the other: a row/column used inside an operation must still be in its original form, so a row changed earlier cannot be used again. $C_1 \to C_1 - C_2$ with $C_2 \to C_2 - C_3$ is fine (do C_1 first); $R_1 \to R_1 - R_2$ with $R_2 \to R_2 - R_1$ is wrong.

Also never multiply the row being changed by a constant without correcting the value ($R_1 \to 2R_1 - R_2$ doubles D).

Grade - **Low**

Practice map, main concept: none

Also used in: Sheet: J-I 2

Four standard determinants

Question 18

Write down all four standard determinants. There are two other common determinants also; write them down.

Model Answer

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3+b^3+c^3-3abc) = -(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

Two other common ones:

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc+ab+bc+ca = abc\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$$

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 19

Are you able to spot the four standard determinants and their results while solving questions?

Model Answer

Take common factors out of rows/columns (and use $R_i \to R_i - R_j$) until a row/column of 1's or the cyclic pattern appears; compare with the list: powers $1, x, x^2, x^3$ in a row $\Rightarrow$ Vandermonde type.

Ex: $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 9 & 16 \end{vmatrix} = (2-3)(3-4)(4-2) = 2$ (standard 2 with $a, b, c = 2, 3, 4$).

Grade - Medium

Practice map, main concept: Sheet: J-I 3; J-M 31; C-I 16; C-II 18 | DPP: 2(1-2, 11, 13, 16); 3(4-5)

Also used in: Sheet: J-I 15; J-II 1, 6, 11; C-I 10; C-II 8, 19 | DPP: 2(4); 3(12); 4(3) | Question Bank: 111

Two Important Operations

Question 20

What are the two important operations and when do we apply them?

Model Answer

(i) A row/column of all 1's (or easily made): subtract it from the others ($R_1 \to R_1 - R_3$, $R_2 \to R_2 - R_3$) and expand along the column of 1's. (ii) If the three rows (columns) add up to a common factor: $R_1 \to R_1 + R_2 + R_3$, take the factor out.

Ex: $\begin{vmatrix} x+1 & 2x & 3 \\ 2x & 3 & x+1 \\ 3 & x+1 & 2x \end{vmatrix} = 0$: each row sums to $3x+4$, so $(3x+4)(\dots) = 0$ and the other factor $3x^2 - 9x + 7 \neq 0$ for real x, giving $x = -\frac{4}{3}$.

Grade - Medium

Practice map, main concept: Sheet: J-II 4 | DPP: 3(10) | Question Bank: 91

Also used in: Sheet: J-II 9 | DPP: 2(10)

Question 21

Which operation do we use when the elements of a row/column are in AP?

Model Answer

If the entries of every row are in AP, $C_1 \to C_1 - 2C_2 + C_3$ gives a zero column, so $D = 0$. If only one row is in AP, use $C_3 \to C_3 - C_2$, $C_2 \to C_2 - C_1$ to make the differences equal.

Ex: $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0$ $(C_1 - 2C_2 + C_3 = 0)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 5; C-II 17, 19 | DPP: 2(3, 6) | Question Bank: 90

Also used in: Sheet: J-A 6 | Question Bank: 113

Multiplication of Determinants

Question 22

Can you multiply any two determinants of the same order in all 4 ways?

Model Answer

Row×column, row×row, column×row, column×column: the (i, j) element is the sum of products of the i-th line of the first with the j-th line of the second.

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Ex: $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = (-2)(-1) = 2 = \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix}$.

Grade - Low

Practice map, main concept: none

Also used in: Sheet: J-I 10-11; J-II 15-18; C-II 11-12 | DPP: 2(12); 4(10)

Question 23

Memorise the result of Question 1 in Multiplication of Determinants.

Model Answer

The determinant of cofactors is D^2 (order 3); in order n it is D^{n-1}:

$$\begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix} = D^2$$

Ex: $D = \text{diag}(1, 2, 3)$ has $D = 6$; its cofactors are $6, 3, 2$ on the diagonal, product $36 = 6^2$. Also $\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = (a^3 + b^3 + c^3 - 3abc)^2$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 9; C-I 1

Also used in: none

Question 24

How do you express a given determinant as the product of two determinants?

Model Answer

Write every element as a sum of products (a dot product of a row of one determinant and a column of another); then the given determinant is that product.

Ex: $(a_i + b_j)^2 = a_i^2 \cdot 1 + a_i \cdot 2b_j + 1 \cdot b_j^2$, so $\left|(a_i + b_j)^2\right|$ = (determinant with rows $a_i^2,\ a_i,\ 1$) $\times$ (determinant with columns $1,\ 2b_j,\ b_j^2$) $= -2(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)(b_1 - b_2)(b_2 - b_3)(b_3 - b_1)$. With $S_r = \alpha^r + \beta^r + \gamma^r$: $|S_{i+j-2}| = [(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)]^2$.

Grade - **Low**

Practice map, main concept: Sheet: J-I 11; J-II 15, 17-18; C-II 11-12 | DPP: 2(12); 3(12); 4(10)

Also used in: none

Question 25

How do you identify whether a question is based on multiplication of determinants?

Model Answer

Each element is a sum of two or three products $(a_1l_1 + b_1l_2 + c_1l_3)$, a square $(a + b)^2$, a power sum $\alpha^n + \beta^n$, $\cos(\alpha_i - \beta_j)$, or the determinant is given with a square (e.g. D^2) or as a product of two determinants.

Ex: $\cos(\alpha_i - \beta_j) = \cos\alpha_i \cos\beta_j + \sin\alpha_i \sin\beta_j$ has only two products, so the 3×3 determinant of these elements is 0 (a third zero column can be added).

Grade - **Low**

Practice map, main concept: Sheet: J-I 10; J-II 16

Also used in: none

System of Equations

Question 26
What do consistent and inconsistent systems mean?

Model Answer
Consistent: at least one solution (unique, or infinitely many). Inconsistent: no solution.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 27
What do trivial and non-trivial solutions mean?

Model Answer
Trivial: every variable is 0 (the origin). Non-trivial: at least one variable is non-zero. They are asked for homogeneous systems (all constants 0), which always have the trivial solution.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 28
Write the condition for a unique solution of two equations in two variables; what does it mean in coordinate geometry?

Model Answer

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

The two lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ intersect at exactly one point.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 29
Write the condition for infinitely many solutions of two equations in two variables; what does it mean in coordinate geometry?

Model Answer

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

The two lines coincide (dependent equations).

Grade - **Low**
Practice map, main concept: Sheet: C-I 14
Also used in: none

Question 30
Write the condition for an inconsistent system of two equations in two variables; what does it mean in coordinate geometry?

Model Answer

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

The two lines are parallel and distinct.
Ex: $x + 2y = 3,\ 2x + 4y = 7$ has no solution ($\frac{1}{2} = \frac{2}{4} \neq \frac{3}{7}$).

Grade - Low

Practice map, main concept: Sheet: J-I 12, 20

Also used in: none

Question 31

Write the condition for a unique solution of three equations in two variables; what does it mean in coordinate geometry?

Model Answer

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

with at least two of the lines not parallel: the three lines are concurrent (meet at one point).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 32

Write the condition for infinitely many solutions of three equations in two variables; what does it mean in coordinate geometry?

Model Answer

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{a_2}{a_3} = \frac{b_2}{b_3} = \frac{c_2}{c_3}$$

All three lines coincide.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 33

Write the condition for an inconsistent system of three equations in two variables; what does it mean in coordinate geometry?

Model Answer

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$$

The three lines have no common point (they form a triangle). Also inconsistent: all three parallel but not all coincident ($D = 0$ then).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 34
What is the condition that three planes form a prism?

Model Answer
The planes meet pairwise (no two parallel) but have no common point: coefficient determinant $D = 0$, the rows of coefficients are not proportional, and the constants do not satisfy the same relation as the rows, so some $D_x, D_y, D_z \neq 0$.

Ex: $x + y + z = 1,\ x + 2y + 3z = 2,\ 2x + 3y + 4z = 5$: $R_3 = R_1 + R_2$ but $5 \neq 1 + 2$, a prism.

Grade - Low

Practice map, main concept: none

Also used in: none

Cramer's Rule

Question 35
Are you able to write all the determinants involved in Cramer's rule?

Model Answer
For $a_1x + b_1y + c_1z = d_1$ etc., D is the coefficient determinant; D_x, D_y, D_z are obtained by replacing the x-, y-, z-column of D by the constants d_1, d_2, d_3:

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \quad D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \quad D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \quad D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 36
Are you able to write Cramer's rule for two equations in two variables?

Model Answer

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}, \quad D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0$$

Ex: $2x + y = 5,\ x - y = 1$: $D = -3,\ D_x = -6,\ D_y = -3 \Rightarrow x = 2,\ y = 1$.

Grade - Low

Practice map, main concept: Sheet: C-II 6

Also used in: none

Question 37
Are you able to write Cramer's rule for three equations in three variables?

Model Answer

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}, \quad z = \frac{D_z}{D} \qquad (D \neq 0)$$

Ex: $x + y + z = 6,\ 2x - y + z = 3,\ x + 2y - z = 2$: $D = 7,\ D_x = 7,\ D_y = 14,\ D_z = 21 \Rightarrow (x, y, z) = (1, 2, 3)$.

Grade - Medium

Practice map, main concept: Sheet: J-II 20; J-M 18; C-II 3, 8

Also used in: Sheet: J-I 13; C-II 5

Question 38

Are you able to draw the chart showing all situations for three equations in three variables?

Model Answer

$D \neq 0$: unique solution (consistent).

$D = 0$ and at least one of $D_x, D_y, D_z \neq 0$: no solution (inconsistent).

$D = D_x = D_y = D_z = 0$: infinitely many solutions or no solution - check further (exception case).

Grade - High

Practice map, main concept: Sheet: J-I 13, 17-19, 22; J-II 22, 24, 29; J-M 2-3, 6-7, 9, 11, 13, 16, 19, 22-23; J-A 1, 5-7; C-I 3, 8, 11; C-II 4-5, 7, 13, 20-22 | DPP: 4(5) | Question Bank: 92-93, 97, 108-110, 112

Also used in: Sheet: J-I 14; J-II 20; J-M 1, 4-5, 10, 15, 17-18, 24-26, 28-29, 32-33, 35

Question 39

What do unique / infinite / no solution represent in 3-D geometry?

Model Answer

Unique: three planes meet at one point. Infinite: three planes pass through a common line (or all coincide). None: three parallel planes, two parallel planes cut by the third, or the planes form a prism.

Grade - Medium

Practice map, main concept: none

Also used in: Question Bank: 98

Question 40

How do you represent/determine infinite solutions of three equations in three variables?

Model Answer

When $D = 0$ find λ, μ with $R_3 = \lambda R_1 + \mu R_2$ (coefficients). The system is consistent (infinite solutions) iff the constants satisfy $d_3 = \lambda d_1 + \mu d_2$; then put $z = t$ and solve the other two equations for x, y.

Ex: $x + 2y + 3z = 4,\ 2x + 3y + 4z = 5,\ 3x + 5y + 7z = k$: $R_3 = R_1 + R_2$, so infinite solutions iff $k = 9$ (no solution otherwise).

Grade - High

Practice map, main concept: Sheet: J-I 14; J-M 1, 4-5, 10, 15, 17, 24-26, 28-29, 32-33, 35 | Question Bank: 98

Also used in: Sheet: J-I 17; J-II 24; J-A 1, 5, 7; C-II 4-5, 7, 13, 20 | Question Bank: 93, 102, 108, 110

Question 41

When should we check the exception case while solving three equations in three variables?

Model Answer

When $D = D_x = D_y = D_z = 0$. Then do not conclude 'infinite': solve two of the equations or compare the planes; three parallel planes (or two parallel) have no solution although all four determinants vanish.

Ex: $2x + y + 4z = 3,\ 2x + y + 4z = 7,\ 2x + y + 4z = 10$: $D = D_x = D_y = D_z = 0$ but no solution.

Grade - Low

Practice map, main concept: none

Also used in: none

System of Homogeneous Linear Equations

Question 42

What do non-trivial solutions of a homogeneous system mean and what is the condition?

Model Answer

A homogeneous system is always consistent ($x = y = z = 0$). It has non-trivial (infinitely many, non-zero) solutions iff $D = 0$; if $D \neq 0$ only the trivial solution exists.

Ex: $x + y + z = 0,\ x + 2y + 3z = 0,\ 2x + 3y + kz = 0$: $D = k - 4$, so non-trivial solutions iff $k = 4$.

Grade - High

Practice map, main concept: Sheet: J-I 15-16, 21; J-II 19, 21, 23, 25; J-M 12, 27, 34; J-A 2; C-I 4-5, 7, 13; C-II 1-2 | DPP: 4(6) | Question Bank: 89, 94, 100, 102, 105, 107, 111

Also used in: none

Miscellaneous (added items)

Question 43

How do you solve an equation of the form (determinant in x) $= 0$?

Model Answer

Simplify with row/column operations (take common factors, create zeros) or use P6 to factor; the roots are the zeros of the factors. Check the number of roots against the degree.

Ex: $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = (x-1)^2(x+2) = 0 \Rightarrow x = 1, 1, -2.$

Grade - Medium

Practice map, main concept: Sheet: J-II 7-9; J-M 14, 21; J-A 3-4; C-I 9, 18, 22 | DPP: 1(1, 17); 2(10); 3(2-3, 7) | Question Bank: 96

Also used in: Sheet: C-II 17

Question 44

How do you find the degree, a coefficient, or the constant term of a determinant that is a polynomial in x?

Model Answer

Constant term $= \Delta(0)$; sum of coefficients $= \Delta(1)$; degree = highest power of x that can occur in a product of one element from each row and column; for an identity in x, put convenient values of x $(0, \pm 1)$ and compare.

Ex: for $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = x^3 - 3x + 2$: degree 3, constant term $\Delta(0) = 2$, coefficient of x is -3.

Grade - Medium

Practice map, main concept: Sheet: J-I 6; C-I 6, 19 | DPP: 2(18); 3(1, 11) | Question Bank: 101

Also used in: Sheet: J-II 25; C-I 9

Question 45

How do you handle a determinant whose elements are trigonometric (or other) functions: simplify, find its range/maximum/minimum or the number of roots?

Model Answer

First reduce with $R_i \to R_i - R_j$ (so common trigonometric parts cancel) and expand; the result is a simple expression such as $a + b\sin 2x$; then use its range $[a - |b|, a + |b|]$ or solve it for the roots in the given interval.

Ex: $\begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix} = 2 + 4\sin 2x$, range $[-2, 6]$.

Grade - High

Practice map, main concept: Sheet: J-II 32; J-M 8; C-I 12, 15, 17; C-II 10, 16, 23-24 | DPP: 4(4, 11) | Question Bank: 95, 103, 106

Also used in: Sheet: J-II 5, 7; J-M 12, 27; J-A 2; C-I 3, 5, 8, 13, 18-19, 22 | DPP: 2(19) | Question Bank: 89